

EXAM 7 – SPRING 2014

15. (1.5 points)

A publicly traded insurer writes three product lines: workers compensation, commercial property, and commercial auto. The following table presents financial data (\$000) for the insurer for 2013.

	<u>Workers Compensation</u>	<u>Commercial Property</u>	<u>Commercial Auto</u>
Earnings	636	318	297
Book Value	8,350	5,860	2,210

Three peer companies have been selected for each product line. These peer companies are monoline insurers that operate in the associated product line only.

<u>Workers Compensation Peer</u>	<u>Price-Earnings Ratio</u>	<u>Commercial Property Peer</u>	<u>Price-Earnings Ratio</u>
Peer 1	21.9	Peer 1	16.4
Peer 2	18.5	Peer 2	19.7
Peer 3	19.6	Peer 3	17.9

<u>Commercial Auto Peer</u>	<u>Price-Earnings Ratio</u>
Peer 1	13.9
Peer 2	17.2
Peer 3	14.8

Calculate the Price to Book Value ratio for the insurer based on relative valuation using multiples.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 15	
TOTAL POINT VALUE: 1.5	LEARNING OBJECTIVE: B3: Value the equity of a firm using comparative or relative valuation methods based on multiples of selected financial variables obtained from either peer companies or from underlying fundamentals.
SAMPLE ANSWERS	
<i>Sample Answer 1 (MEAN):</i> Step 1: Calculate average P-E ratios based on peer companies by product line WC: P-E ratio = $(21.9 + 18.5 + 19.6) / 3 = 20.0$ CP: P-E ratio = $(16.4 + 19.7 + 17.9) / 3 = 18.0$ CA: P-E ratio = $(13.9 + 17.2 + 14.8) / 3 = 15.3$ Step 2: Calculate the price of each segment for the company. Price = P-E Ratio $\times$ Latest Fiscal Year Earnings WC: Price = $20.0 \times 636 = \\$12,720K$ CP: Price = $18.0 \times 318 = \\$5,724K$ CA: Price = $15.3 \times 297 = \\$4,544K$ Step 3: Calculate total price of the company Price = $12,720 + 5,724 + 4,544 = \\$22,988K$ Step 4: Calculate Total Book Value Total Book value = $8,350 + 5,860 + 2,210 = 16,420$ Step 5: Calculate Price to Book Value Ratio Price to Book Value Ratio = $22,988 / 16,420 = 1.40$ <i>Sample Answer 2 (MEDIAN):</i> Step 1: Select median P-E ratio of peer companies by product line WC: P-E ratio = 19.6 CP: P-E ratio = 17.9 CA: P-E ratio = 14.8	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Step 2: Calculate the price of each segment for the company.

Price = P-E Ratio $\times$ Latest Fiscal Year Earnings

WC: Price = $19.6 \times 636 = \$12,466K$

CP: Price = $17.9 \times 318 = \$5,692K$

CA: Price = $14.8 \times 297 = \$4,396K$

Step 3: Calculate total price of the company

Price = $12,466 + 5,692 + 4,396 = \$22,554K$

Step 4: Calculate Total Book Value

Total Book value = $8,350 + 5,860 + 2,210 = 16,420$

Step 5: Calculate Price to Book Value Ratio

Price to Book Value Ratio = $22,554 / 16,420 = 1.37$

EXAMINER'S REPORT

Candidates performed very well on average. Candidates needed to know how to value the firm using financial information from peer companies. Some candidates calculated the P-BV of each 3 lines separately and then summed them to get to the company P-BV. A few others had difficulty figuring out how to use the data given (P/E ratios of competitors) to get to the P/BV of the firm.